Chapter S

Additional files

This document holds the tables and figures that support the main chapters but are too long to include in them. They are grouped by the chapter they support and numbered in order as Additional files.

S.1 Noise robustness across molecular representations and model architectures

Additional file 1: Hyperparameters for every model configuration, and the descriptors that make up PDV

Table A lists the default settings of every model configuration, Table B the tuned settings that replace a default on each dataset, and Table C the 200 descriptors that make up PDV.

Table S.1: **Table A: default hyperparameters for every model configuration in the study.** Every value is read from `models/model_defaults.py` at spec version 1.7.0, spec hash `9249926c38dc`, which is the file both the QM9 pipeline and the assay pipeline load their settings from. Every QM9 results row carries the spec version and the spec hash as two of its columns, so a QM9 table can be traced back to the settings that produced it; the three assay datasets are written by a second pipeline whose results files do not carry those two columns. The configuration list is the model roster the job generator submits. One setting is used for every representation and for every dataset, except where Table B gives a tuned replacement. The support vector machine uses an RBF kernel on every representation, so no result in this study carries a kernel that changes with the representation. Two values are not held in the shared spec and name their source file in the third column.

Configuration	**Hyperparameter**	**Default value**
Shared by every neural network below		
Training, all networks	optimizer	adam
	lr	0.001
	weight_decay	0.0
	batch_size	32
	epochs	100 cap; early stopping usually reaches it first
	patience	10 epochs without a strictly better mean validation loss
	restore_best_weights	True the reported fit is the best epoch, not the last
	val_loss_reduction	mean
	improvement_tolerance	0.0
	mc_passes	100 stochastic forward passes at inference
	standardise_targets	True fitted on training labels alone
Tree and boosting models		
RF `rf`	n_estimators	100

Configuration	Hyperparameter	Default value
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
RF300 `rf300`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
XGBoost `xgboost`	n_estimators	100
	max_depth	6
	learning_rate	0.1
	subsample	1.0
	colsample_bytree	1.0
	colsample_bylevel	1.0
	min_child_weight	1
	gamma	0.0
	reg_alpha	0.0
	reg_lambda	1.0
LightGBM `lgb`	n_estimators	100
	learning_rate	0.1
	num_leaves	31
	max_depth	-1 no limit
	subsample	1.0
	colsample_bytree	1.0
	min_child_samples	20
	reg_alpha	0.0

Configuration	Hyperparameter	Default value
	reg_lambda	0.0
NGBoost `ngboost`	n_estimators	500
	learning_rate	0.01
	natural_gradient	True
	dist	Normal ngboost.distns.Normal
	score	MLE ngboost.scores.MLE, the log scoring rule
	minibatch_frac	1.0
	col_sample	1.0
	base_max_depth	3 depth of one boosting stage
	base_criterion	friedman_mse
	early_stopping_rounds	50 stages without a better validation log likelihood
	use_best_iteration	True predictions use the best stage, not every stage fitted
	ngboost_ensemble_seeds	1 1 = one fit, no seed ensemble
QRF `qrf`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
Kernel models		
SVM `svm`	C	1.0

Configuration	Hyperparameter	Default value
	gamma	scale
	kernel	rbf
GP `gauche_rbf`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median

Configuration	Hyperparameter	Default value
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Tanimoto `gauche` run on ecfp4 only	kernel	tanimoto set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default

Configuration	Hyperparameter	Default value
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Hetero `heteroscedastic_gp`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime

Configuration	Hyperparameter	Default value
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
	noise network hidden sizes	[64, 64] models/models.py, not in the shared spec
	noise network output	Softplus, plus 10^{-4} models/models.py, not in the shared spec
	Adam learning rate, process	0.1 models/models.py, not in the shared spec
	Adam learning rate, noise network	0.001 models/models.py, not in the shared spec
	noise network epochs	100 the shared epoch cap
Neural networks		
NN-α `dnn`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu

Configuration	Hyperparameter	Default value
NN-β	hidden_size	128
`mlp`		
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
BNN-α	hidden_sizes	[128, 64]
`dnn_bnn_full`		
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo
		1 / number of training molecules
BNN-β	hidden_size	128
`mlp_bnn_full`		
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo
		1 / number of training molecules
VBLL-α	hidden_sizes	[128, 64]
`dnn_bnn_full_variational`		
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-β	hidden_size	128
`mlp_bnn_full_variational`		

Configuration	Hyperparameter	Default value
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-α (noise head) `dnn_bnn_full_variational_hetero`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
VBLL-β (noise head) `mlp_bnn_full_variational_hetero`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
BNN-α (variance head) `dnn_bnn_full_mve`	hidden_sizes	[128, 64]
	dropout_rate	0.2

Configuration	Hyperparameter	Default value
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec
BNN-β (variance head) `mlp_bnn_full_mve`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec

Table S.2: **Table B: the tuned settings that replace a default, and the four datasets they apply to.** A setting is chosen once per model per dataset, not per representation, and is then used for all six representations: a two-way comparison across model and representation cannot be read if each cell carries its own hyperparameters. Only the Bayesian and variational networks are tuned; every other configuration in Table A trains at the default. A model with no row for a dataset kept the default on that dataset. Learning rates are printed to three significant figures; the source files hold the full value. Source files: `results/master_tuned_hyperparameters.json` for QM9 and `results/master_tuned_hyperparameters_lab.json` for the three assay datasets.

Dataset	Configuration	Hyperparameter	Tuned value
QM9	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	VBLL-β	dropout_rate	0.357
		hidden_size	64
		lr	0.00119
		num_hidden_layers	1
Caco-2	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	BNN-α	activation	tanh

Dataset	Configuration	Hyperparameter	Tuned value
		hidden_size1	64
		hidden_size2	64
Caco-2	VBLL-α	activation	relu
		hidden_size1	64
		hidden_size2	32
Caco-2	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
Caco-2	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	NN-α	activation	tanh
		hidden_size1	64

Dataset	Configuration	Hyperparameter	Tuned value
		hidden_size2	64
LogD	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	VBLL-α	activation	tanh
		hidden_size1	256
		hidden_size2	32
LogD	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	VBLL-β	dropout_rate	0.079
		hidden_size	128
		lr	0.00011
		num_hidden_layers	2

Table S.3: **Table C: the 200 descriptors that make up the PDV representation.** One entry is one feature of the vector, in the order the pipeline computes them. The list is fixed in the code rather than taken from RDKit's current default set, which grows between releases, so the same 200 features are built on every dataset and in every run. They were computed with RDKit 2022.09.5. Descriptors that return a non-finite value on a molecule have that value replaced by zero before the standardization constants are computed. Source: `DEFAULT_DESCRIPTOR_LIST` in `scripts/process_and_train.py`.

BalabanJ	NOCount	SlogP_VSA7	fr_bicyclic
BertzCT	NumAliphaticCarbocycles	SlogP_VSA8	fr_diazo
Chi0	NumAliphaticHeterocycles	SlogP_VSA9	fr_dihydropyridine
Chi0n	NumAliphaticRings	TPSA	fr_epoxide
Chi0v	NumAromaticCarbocycles	VSA_EState1	fr_ester
Chi1	NumAromaticHeterocycles	VSA_EState10	fr_ether

Chi1n	NumAromaticRings	VSA_EState2	fr_furan
Chi1v	NumHAcceptors	VSA_EState3	fr_guanido
Chi2n	NumHDonors	VSA_EState4	fr_halogen
Chi2v	NumHeteroatoms	VSA_EState5	fr_hdrzine
Chi3n	NumRadicalElectrons	VSA_EState6	fr_hdrzone
Chi3v	NumRotatableBonds	VSA_EState7	fr_imidazole
Chi4n	NumSaturatedCarbocycles	VSA_EState8	fr_imide
Chi4v	NumSaturatedHeterocycles	VSA_EState9	fr_isocyan
EState_VSA1	NumSaturatedRings	fr_Al_COO	fr_isothiocyan
EState_VSA10	NumValenceElectrons	fr_Al_OH	fr_ketone
EState_VSA11	PEOE_VSA1	fr_Al_OH_noTert	fr_ketone_Topliss
EState_VSA2	PEOE_VSA10	fr_ArN	fr_lactam
EState_VSA3	PEOE_VSA11	fr_Ar_COO	fr_lactone
EState_VSA4	PEOE_VSA12	fr_Ar_N	fr_methoxy
EState_VSA5	PEOE_VSA13	fr_Ar_NH	fr_morpholine
EState_VSA6	PEOE_VSA14	fr_Ar_OH	fr_nitrile
EState_VSA7	PEOE_VSA2	fr_COO	fr_nitro
EState_VSA8	PEOE_VSA3	fr_COO2	fr_nitro_arom
EState_VSA9	PEOE_VSA4	fr_C_O	fr_nitro_arom_nonortho
ExactMolWt	PEOE_VSA5	fr_C_O_noCOO	fr_nitroso
FpDensityMorgan1	PEOE_VSA6	fr_C_S	fr_oxazole
FpDensityMorgan2	PEOE_VSA7	fr_HOCCN	fr_oxime
FpDensityMorgan3	PEOE_VSA8	fr_Imine	fr_para_hydroxylation
FractionCSP3	PEOE_VSA9	fr_NH0	fr_phenol
HallKierAlpha	RingCount	fr_NH1	fr_phenol_noOrthoHbond
HeavyAtomCount	SMR_VSA1	fr_NH2	fr_phos_acid
HeavyAtomMolWt	SMR_VSA10	fr_N_O	fr_phos_ester
Ipc	SMR_VSA2	fr_Ndealkylation1	fr_piperdine
Kappa1	SMR_VSA3	fr_Ndealkylation2	fr_piperzine
Kappa2	SMR_VSA4	fr_Nhpyrrole	fr_priamide
Kappa3	SMR_VSA5	fr_SH	fr_prisulfonamd
LabuteASA	SMR_VSA6	fr_aldehyde	fr_pyridine
MaxAbsEStateIndex	SMR_VSA7	fr_alkyl_carbamate	fr_quatN
MaxAbsPartialCharge	SMR_VSA8	fr_alkyl_halide	fr_sulfide
MaxEStateIndex	SMR_VSA9	fr_allylic_oxid	fr_sulfonamd
MaxPartialCharge	SlogP_VSA1	fr_amide	fr_sulfone

MinAbsEStateIndex	SlogP_VSA10	fr_amidine	fr_term_acetylene
MinAbsPartialCharge	SlogP_VSA11	fr_aniline	fr_tetrazole
MinEStateIndex	SlogP_VSA12	fr_aryl_methyl	fr_thiazole
MinPartialCharge	SlogP_VSA2	fr_azide	fr_thiocyan
MolLogP	SlogP_VSA3	fr_azo	fr_thiophene
MolMR	SlogP_VSA4	fr_barbitur	fr_unbrch_alkane
MolWt	SlogP_VSA5	fr_benzene	fr_urea
NHOHCount	SlogP_VSA6	fr_benzodiazepine	qed

Additional file 2: Configurations excluded from the robustness analysis

For each model and representation on the QM9 HOMO–LUMO gap, the number of noise strategies for which the configuration was excluded because its clean-label R^2 was too low.

Table S.4: **Configurations excluded from robustness analysis due to baseline $R^2 \leq 0.6$ on the QM9 HOMO–LUMO gap, summarized by model and representation.** Each cell shows the number of noise strategies (out of 6) for which the configuration was excluded.

Model	Mol2Vec	MHGGNN	SMILES	Rand. SMILES
BNN-α	6	6	—	—
BNN-β	6	6	—	—
VBLL-α	6	6	—	—
VBLL-β	6	6	—	—
RF	—	—	—	4
QRF	—	—	—	4
NGBoost	—	—	6	4

Additional file 3: Spearman correlations between the robustness profiles of the models

Pairs of models whose noise degradation slope profiles on the QM9 HOMO–LUMO gap correlate at $\rho \geq 0.95$, and which model of each pair was left out of the ANOVA.

Table S.5: **Pairwise Spearman rank correlations between model noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap ($\rho \geq 0.95$).** "Model A" and "Model B" label each pair, both treated symmetrically. Checkmarks indicate pairs where one model was excluded from the ANOVA due to near-redundancy. Gauche (RBF) pairs omitted (all $\rho = 1.0$, $n = 6$, PDV-only artifact).

Model A	**Model B**	n	ρ	**Excluded**
VBLL-β	NGBoost	18	0.998	
LightGBM	XGBoost	46	0.998	
Gauche	VBLL-β	18	0.996	✓
QRF	RF	40	0.995	✓
VBLL-α	LightGBM	24	0.992	
RF	XGBoost	46	0.989	
NGBoost	XGBoost	40	0.988	
NGBoost	RF	40	0.987	
LightGBM	RF	46	0.986	
VBLL-β	SVM	24	0.986	
QRF	XGBoost	40	0.986	✓
NGBoost	QRF	34	0.986	✓
VBLL-α	NGBoost	18	0.986	
VBLL-α	XGBoost	24	0.984	
LightGBM	NGBoost	40	0.984	
LightGBM	QRF	40	0.984	✓
VBLL-α	VBLL-β	24	0.983	
NN-α	NN-β	52	0.983	
LightGBM	SVM	46	0.982	
BNN-β	SVM	34	0.981	
VBLL-α	SVM	24	0.979	
BNN-α	VBLL-β	24	0.979	
VBLL-α	QRF	24	0.978	✓
SVM	XGBoost	46	0.978	
BNN-α	BNN-β	36	0.977	
BNN-α	SVM	34	0.976	
VBLL-α	Gauche	18	0.975	✓
VBLL-α	RF	24	0.975	
QRF	SVM	40	0.974	✓
BNN-α	VBLL-α	24	0.974	
LightGBM	VBLL-β	24	0.973	
VBLL-α	BNN-β	24	0.972	
Gauche	SVM	40	0.972	✓
Gauche	XGBoost	44	0.970	✓
NGBoost	SVM	40	0.970	
NN-α	BNN-β	36	0.970	
Gauche	LightGBM	42	0.969	✓
RF	SVM	46	0.969	
BNN-β	VBLL-β	24	0.968	
VBLL-β	QRF	24	0.966	✓
NN-α	BNN-α	36	0.965	
VBLL-β	XGBoost	24	0.965	
NN-β	SVM	46	0.962	
NN-α	SVM	46	0.960	
Gauche	NGBoost	34	0.959	✓

Model A	Model B	n	ρ	Excluded
LightGBM	BNN-β	36	0.959	
BNN-α	NN-β	36	0.958	
BNN-α	QRF	28	0.956	✓
Gauche	QRF	34	0.956	✓
VBLL-β	RF	24	0.955	
Gauche	RF	40	0.954	✓
BNN-β	XGBoost	34	0.953	
BNN-β	QRF	28	0.952	✓
NN-α	VBLL-β	24	0.952	
NN-β	BNN-β	36	0.952	
BNN-α	LightGBM	36	0.951	
NN-β	VBLL-β	24	0.950	
BNN-α	Gauche	30	0.950	✓

Additional file 4: Spearman correlations between the robustness profiles of the representations

Pairwise correlations between the noise degradation slope profiles of the representations on the QM9 HOMO–LUMO gap, and which representation of each pair was left out of the ANOVA.

Table S.6: **Pairwise Spearman rank correlations between representation noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap.** "Rep A" and "Rep B" label each pair of molecular representations, both treated symmetrically. Checkmarks indicate pairs where one representation was excluded from the ANOVA.

Rep. A	Rep. B	n	ρ	Excluded
ECFP4	Morgan	60	0.995	✓
MHGGNN	Morgan	48	0.994	✓
Rand. SMILES	SMILES	16	0.994	✓
Rand. SMILES	SNS	16	0.994	✓
ECFP4	MHGGNN	54	0.993	
ECFP4	SNS	56	0.992	✓
Morgan	SNS	52	0.992	✓
MHGGNN	SNS	44	0.991	✓
PDV	ECFP4	72	0.990	
PDV	MHGGNN	48	0.989	
SMILES	SNS	52	0.988	✓
Morgan	Rand. SMILES	16	0.988	✓
Morgan	SMILES	54	0.988	✓
PDV	Morgan	54	0.986	✓
PDV	Rand. SMILES	12	0.986	✓
PDV	SNS	50	0.984	✓
ECFP4	SMILES	72	0.984	
Mol2Vec	Morgan	48	0.982	✓
MHGGNN	Mol2Vec	54	0.982	
ECFP4	Mol2Vec	54	0.982	

Rep. A	**Rep. B**	n	ρ	**Excluded**
PDV	PDV (binary)	72	0.980	✓
MHGGNN	SMILES	48	0.980	
PDV	Mol2Vec	48	0.979	
ECFP4	Rand. SMILES	16	0.979	✓
PDV	SMILES	66	0.978	
MHGGNN	PDV (binary)	54	0.972	✓
Mol2Vec	SNS	44	0.969	✓
Mol2Vec	SMILES	48	0.966	
ECFP4	PDV (binary)	78	0.965	✓
Morgan	PDV (binary)	60	0.962	✓
PDV (binary)	SNS	56	0.961	✓
Mol2Vec	PDV (binary)	54	0.956	✓
MHGGNN	Rand. SMILES	16	0.956	✓
PDV (binary)	SMILES	72	0.941	✓
PDV (binary)	Rand. SMILES	16	0.929	✓
Mol2Vec	Rand. SMILES	16	0.891	✓

Additional file 5: Intraclass correlation coefficients for model pairs

ICC(1,1) of the noise degradation slope profiles for the model pairs with ICC ≥ 0.5, on the QM9 HOMO–LUMO gap.

Table S.7: **Intraclass correlation coefficients (ICC(1,1)) for model pairs with ICC ≥ 0.5, assessing within-family consistency of noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap.** ICC(1,1) measures the proportion of total variance attributable to between-configuration differences; values near 1.0 indicate that two models rank configurations almost identically. Lower (more negative) NDS indicates faster degradation under noise. The Family column groups each pair: "Tree" (both tree-based), "α/β" (architecture-pair partners such as BNN-α/BNN-β), "α" (both share the α neural-network backbone), or "Cross" (cross-family pair).

Model A	**Model B**	n	**ICC(1,1)**	**Mean $\lvert\Delta$NDS$\rvert$**	**Family**
LightGBM	XGBoost	8	0.944	0.0060	Tree
BNN-α	BNN-β	6	0.846	0.0101	α/β
QRF	RF	7	0.810	0.0216	Tree
VBLL-α	VBLL-β	4	0.782	0.0108	α/β
QRF	XGBoost	7	0.754	0.0157	Tree
LightGBM	QRF	7	0.743	0.0133	Tree
VBLL-α	LightGBM	4	0.698	0.0115	Cross
VBLL-β	SVM	4	0.689	0.0133	Cross
RF	XGBoost	8	0.678	0.0212	Tree
VBLL-α	XGBoost	4	0.656	0.0133	Cross
BNN-α	VBLL-α	4	0.604	0.0134	α
LightGBM	RF	8	0.586	0.0244	Tree
LightGBM	VBLL-β	4	0.577	0.0174	Cross
LightGBM	SVM	8	0.561	0.0117	Cross
BNN-α	VBLL-β	4	0.552	0.0123	α/β
VBLL-β	XGBoost	4	0.544	0.0195	Cross
VBLL-α	QRF	4	0.540	0.0229	Cross
VBLL-α	SVM	4	0.522	0.0124	Cross
Gauche	VBLL-β	3	0.519	0.0113	Cross

Additional file 6: Global overview of noise robustness on ECFP4

Noise robustness of every model on the ECFP4 representation, on the QM9 HOMO–LUMO gap.

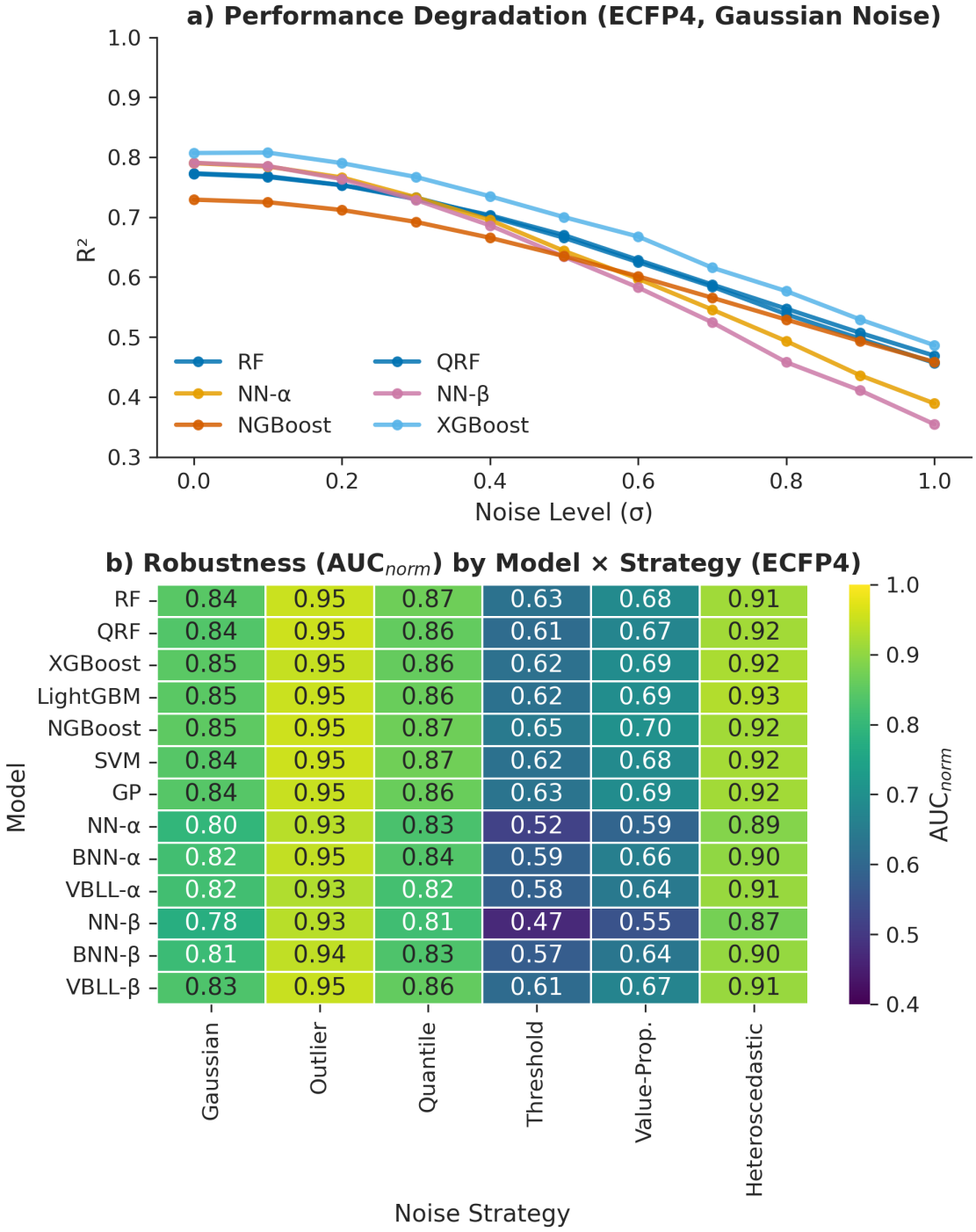


Figure S.1: **Global overview of noise robustness on the ECFP4 representation, evaluated on QM9 HOMO–LUMO gap.** a) Performance degradation curves (R^2 versus noise level σ) under Gaussian noise. b) Noise degradation slope (NDS) heatmap across model architectures and noise strategies; lower (more negative) NDS indicates faster degradation under noise.

Additional file 7: Clean R^2 against robustness on PDV under Gaussian noise

Clean-label R^2 against the noise degradation slope for every model on PDV under

Gaussian noise, on the QM9 HOMO–LUMO gap.

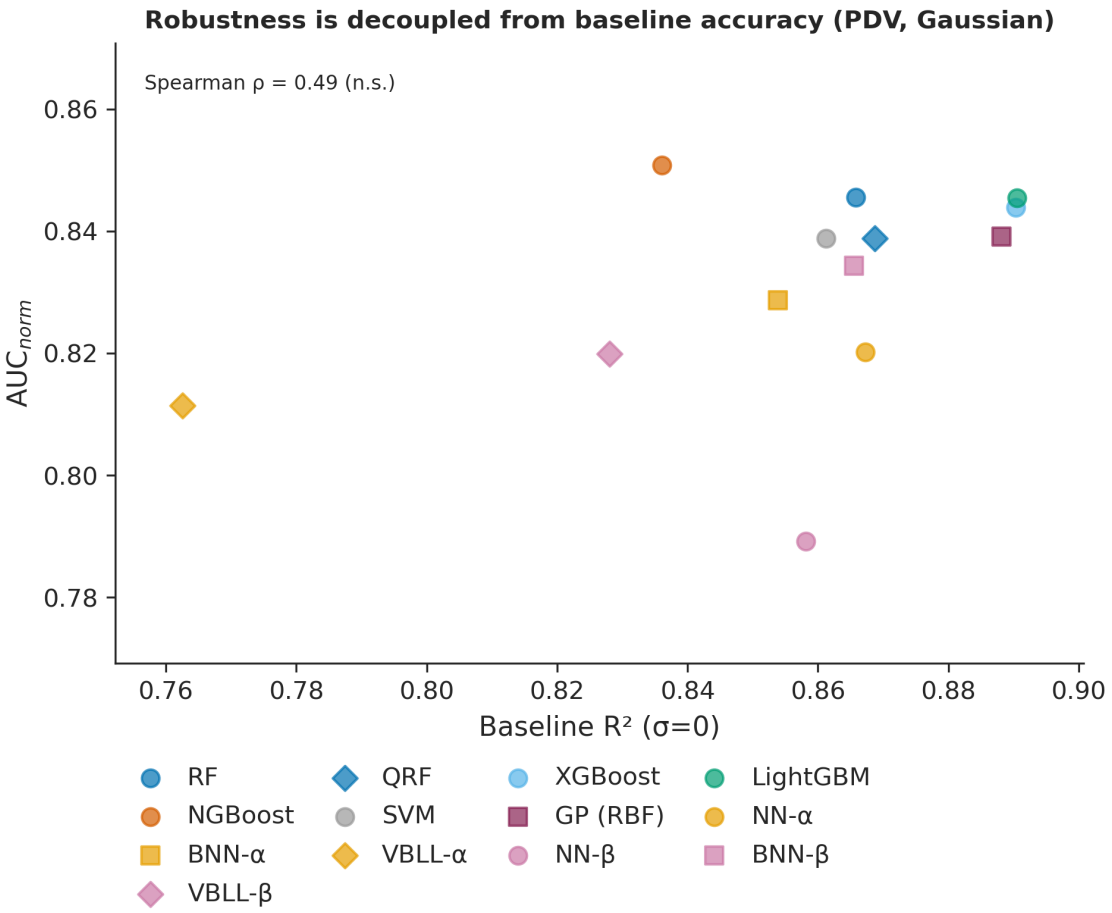


Figure S.2: **Baseline R^2 versus noise degradation slope (NDS) scatter for all models on the physicochemical descriptor vector (PDV) representation under Gaussian noise, on the QM9 HOMO–LUMO gap.** Lower (more negative) NDS indicates faster performance loss under noise.

Additional file 8: Bayesian transformation effects by representation

The effect of each Bayesian transformation of the neural networks on the noise degradation slope, by representation, on the QM9 HOMO–LUMO gap.

Table S.8: **Neural network Bayesian transformation effects on noise degradation slope (NDS) by representation, on the QM9 HOMO–LUMO gap.** Positive Δ NDS indicates the Bayesian variant is more robust (less negative slope) than its deterministic counterpart.

Model	Representation	Baseline R^2	NN-β NDS	BNN-β NDS	Δ NDS
NN-β	ECFP4	0.791	−0.464	−0.388	+0.076
NN-β	PDV	0.858	−0.487	−0.382	+0.106
NN-β	SMILES	0.721	−0.558	−0.365	+0.193

Additional file 9: Uncertainty metrics for probabilistic models on ECFP4

Uncertainty metrics of the probabilistic models on ECFP4 under Gaussian noise, on the QM9 HOMO–LUMO gap.

Table S.9: **Uncertainty quantification metrics for probabilistic models on the ECFP4 representation (Gaussian noise strategy) on the QM9 HOMO–LUMO gap.** Unc-Error ρ: Spearman correlation between predicted uncertainty and absolute prediction error. Unc-Noise ρ: Spearman correlation between predicted uncertainty and injected noise level σ. ECE: expected calibration error (lower = better-calibrated). Coverage targets: 68% (1σ), 95% (2σ).

Model	**Unc-Error ρ**	**Unc-Noise ρ**	**ECE**	**Cov. 1σ**	**Cov. 2σ**
QRF	**0.25**	0.27	0.19	74.7%	94.6%
NGBoost	0.22	**0.37**	**0.11**	69.2%	94.2%
GP	0.18	0.34	0.23	78.6%	96.0%
BNN-β	0.20	0.37	0.29	78.6%	95.2%
BNN-α	0.15	0.32	0.24	76.6%	95.1%
VBLL-β	0.19	0.34	0.19	73.9%	95.5%
VBLL-α	0.11	0.31	0.22	74.5%	95.1%

Additional file 10: Variance decomposition on the three assay datasets

ANOVA variance decomposition (η^2, in percent) on LogD, Caco-2 efflux and hERG K_i.

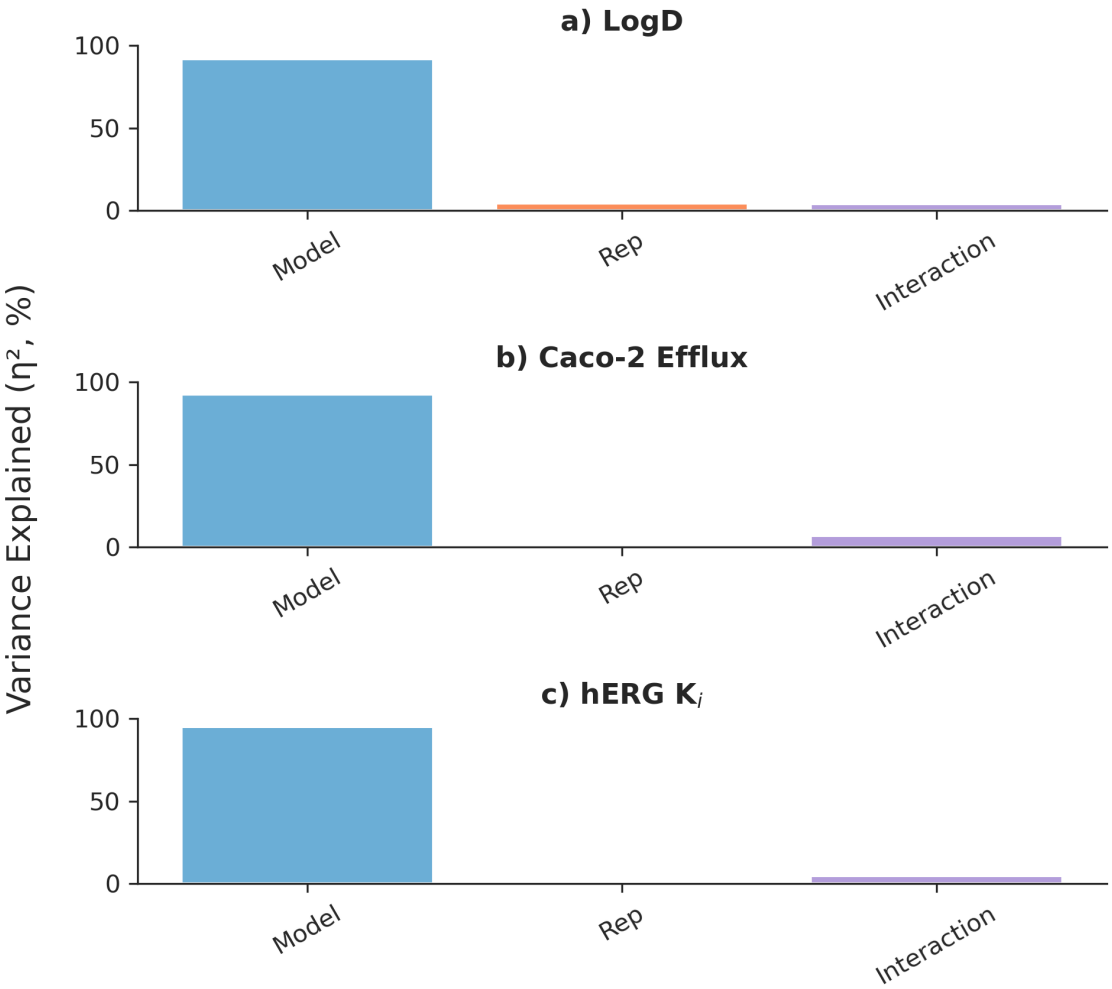


Figure S.3: **ANOVA variance decomposition (η^2, %) on external validation datasets: a) LogD, b) Caco-2 Efflux, and c) hERG K$_i$.** Model, representation, and interaction contributions to noise degradation slope (NDS) variance for each dataset; lower (more negative) NDS indicates faster performance loss under noise.

Additional file 11: Random forest against quantile regression forest on the three assay datasets

The noise degradation slope of random forest and of quantile regression forest on LogD, Caco-2 efflux and hERG K_i.

Table S.10: **Random forest (RF) vs quantile regression forest (QRF) noise degradation slope (NDS) on the three external validation datasets (LogD, Caco-2 Efflux, hERG K_i).** Lower (more negative) NDS indicates faster degradation under noise.

Dataset	**RF NDS**	**QRF NDS**	**Δ NDS**
ChEMBL-hERG-Ki	−0.231	−0.359	−0.128
OpenADMET-Caco2	−0.534	−0.627	−0.094
OpenADMET-LogD	−0.101	−0.185	−0.083

S.2 The flaws of the similarity assumption

One row of Table S.11 is one representation on one dataset. Representations are listed in each dataset's pool order.

Table S.11: All-pairs correlation, five-nearest-neighbor prediction and activity cl for every representation in each regression dataset's pool. All pairs: Spearm correlation between pair distance and absolute label difference, on a fixed sam of 2,000 molecules (seed 0) where a dataset has more. Five neighbors: Spearm correlation and RMSE (in the label's units) of the mean label of the five near molecules from other scaffold groups. Cliff molecules: molecules in at least one this representation's closest pairs whose labels differ by more than the dataset's g (Table 4.4). The number of closest pairs is the number ECFP4 has at Tanimo ≥ 0.55, a few more where distances tie at the cut-off. Shared: percentage of ECFP similar pairs that are also among this representation's closest pairs. HLM clearanc pool has no ECFP4.

		All pairs	Five neighbors		
Dataset	Representation	Spearman	Spearman	RMSE	m
hERG K_i	MHG-GNN	0.07	0.71	0.65	
	Mordred	0.07	0.72	0.62	
	WL graph kernel	0.04	0.71	0.64	
	Avalon	0.02	0.69	0.65	
	GROVER	0.05	0.70	0.63	
	PDV	0.08	0.73	0.62	
	ECFP4	0.06	0.72	0.63	
	Atom pairs	0.04	0.74	0.60	
	ChemBERT	0.07	0.68	0.66	

Dataset	Representation	All pairs	5-NN Spearman	5-NN RMSE	Cliff m
	SMI-TED	0.09	0.69	0.65	
LogD	MHG-GNN	0.09	0.81	0.66	
	Mordred	0.06	0.80	0.68	
	GROVER	0.09	0.81	0.67	
	ECFP4	0.12	0.76	0.74	
	SMI-TED	0.10	0.76	0.74	
	MoLFormer	0.13	0.77	0.73	
	Avalon	0.04	0.73	0.77	
	Topological torsions	0.10	0.74	0.76	
	ChemBERT	0.09	0.75	0.75	
	GraphMVP	0.11	0.76	0.74	
Caco-2 efflux	WL graph kernel	0.13	0.58	0.35	
	MHG-GNN	0.16	0.60	0.34	
	ECFP4	0.08	0.59	0.34	
	PDV	0.18	0.62	0.34	
	GROVER	0.06	0.61	0.34	
	Avalon	−0.00	0.56	0.35	
	Topological torsions	0.15	0.61	0.34	
	SMI-TED	0.14	0.57	0.35	
	Atom pairs	0.08	0.59	0.34	
	MoLFormer	0.09	0.59	0.34	
RLM clearance	PDV	0.06	0.56	0.62	
	MHG-GNN	0.02	0.47	0.67	
	SMI-TED	0.04	0.46	0.67	
	Mordred	0.07	0.49	0.66	
	MoLFormer	0.08	0.44	0.68	
	GROVER	0.05	0.43	0.69	
	WL graph kernel	0.00	0.40	0.71	
	Atom pairs	0.07	0.47	0.67	
	Avalon	0.04	0.29	0.74	
	ECFP4	0.02	0.36	0.71	
MDR1–MDCK efflux	Mordred	0.16	0.58	0.57	

Dataset	Representation	All pairs	5-NN Spearman	5-NN RMSE	Cliff m
	PDV	0.23	0.64	0.54	
	MHG-GNN	0.20	0.52	0.62	
	Avalon	−0.04	0.50	0.61	
	GROVER	−0.03	0.47	0.63	
	SMI-TED	0.10	0.59	0.56	
	Atom pairs	0.00	0.58	0.57	
	WL graph kernel	0.17	0.47	0.64	
	MACCS keys	−0.04	0.54	0.59	
	ECFP4	0.02	0.51	0.61	
HLM clearance	Mordred	0.07	0.49	0.55	
	PDV	0.10	0.55	0.52	
	MHG-GNN	0.08	0.46	0.56	
	SMI-TED	0.03	0.45	0.56	
	MoLFormer	0.00	0.43	0.56	
	GROVER	−0.03	0.41	0.59	
	Atom pairs	−0.04	0.47	0.55	
	ChemBERT	−0.02	0.39	0.58	
	WL graph kernel	0.12	0.38	0.59	
	Avalon	0.01	0.32	0.61	
QM9	Mordred	0.18	0.84	0.68	
	Avalon	0.19	0.83	0.70	
	SMI-TED	0.13	0.85	0.65	
	ECFP4	0.19	0.88	0.64	
	SELFormer	0.20	0.82	0.73	
	ChemBERTa-2	0.19	0.89	0.61	
	Mol2vec	0.14	0.86	0.66	
	ChemBERT	0.19	0.83	0.72	
	GraphMVP	0.20	0.83	0.71	
	MoLFormer	0.21	0.84	0.70	

S.3 Building adaptive hybrid representations

Additional file 12: Settings of every model, as downstream model and as internal evaluator

One row is one model. The middle two columns give its settings when it scores a hybrid (downstream model) and when it scores one representation inside the search (internal evaluator). The models fitted only to compute feature importance are described in Chapter 5.

Table S.12: Settings of every model. p is the number of features. "Molecules" is the number of training molecules the internal evaluator is fitted on; above it, a random subset of that size is drawn with a fixed seed. Settings not listed are the library's defaults. Every setting is the same on every dataset. The Gaussian process runs only on the datasets named in Chapter 5, "Models and metrics".

Model	As downstream model	As internal evaluator	Molecules	Library; source
Random forest	100 trees, no depth limit; $\sqrt{p}$ features tried at each split	50 trees, depth at most 10; all p features tried at each split on regression, $\sqrt{p}$ on classification	all	scikit-learn [1]; Breiman [2]
LightGBM	100 trees	50 trees, depth at most 4	all	lightgbm; Ke et al. [3]
XGBoost	100 trees; half of the features sampled for each tree	50 trees, depth at most 4	all	xgboost; Chen and Guestrin [4]
SVM	RBF kernel, $C = 1$. Not fitted above 10,000 molecules, or above 5,000 under uncapped augmentation	RBF kernel, $C = 1$	at most 500	scikit-learn; Vapnik [5]
Gaussian process	Regression: RBF plus white-noise kernel, labels normalized, 3 optimizer restarts. Classification: RBF kernel. Not fitted above 1,000 molecules	Regression: scaled RBF kernel, fixed noise variance 0.01, no restarts. Classification: scaled RBF kernel, no restarts	at most 500	scikit-learn; Rasmussen and Williams [6]

Model	As downstream model	As internal evaluator	Molecules	Library; source
DNN	Hidden layers of 256, 128 and 64 units, ReLU, dropout 0.2 [7]; Adam [8] at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules that share no group with the rest, using the outer split's grouping; labels z-normalized	The same, except that the early-stopping 20% is drawn at random	all	PyTorch [9]

Additional file 13: Downstream models left out, by dataset and augmentation setting

One row is one rule that leaves a downstream model out. Only the 16 datasets of Table 5.9 are listed; a dataset and setting not named in any row runs every downstream model in Table 5.7.

Table S.13: Downstream models left out, by dataset and augmentation setting. Every rule was set so that a search finishes within the time limit of one job; none was set from scores. hERG K_i drops no model under any setting.

Model left out	Augmentation setting	Datasets
XGBoost	all three	H3K4me3, TATA promoter
SVM	all three	datasets above 10,000 molecules: H3K4me3
SVM	uncapped	datasets above 5,000 molecules: QM9, LogD[a], TATA promoter[a]
SVM	uncapped	Koenig expression
SVM	capped	Caco-2 efflux, LogD, rat liver microsome clearance, MDR1–MDCK efflux, Buchwald–Hartwig plates

Model left out	Augmentation setting	Datasets
Random forest and SVM	capped	QM9
Random forest	uncapped	QM9, LogD
Gaussian process	all three	every dataset except humanness (Chapter 5, "Models and metrics")

[a]Counted on the whole dataset: LogD has 5,028 molecules and TATA promoter 5,274. A search run inside one outer fold is given about 80% of them and keeps the SVM.

Additional file 14: Size of the search

One row is one downstream model. The counts are per dataset, objective and augmentation setting, with adaptive pruning off.

Table S.14: Number of configurations searched per dataset, objective and augmentation setting. A configuration is one choice of downstream model, internal evaluator, allocation method, importance method, representation count and budget. At the full tier each (downstream model, internal evaluator) pair runs 5 allocation methods × 3 importance methods × 4 representation counts × 8 budgets = 480 configurations; at the fast tier it runs $3 \times 1 \times 2 \times 8 = 48$ (Table 5.7). "Matched" is counted once where it repeats an evaluator already in the list. Each configuration is scored on 5 folds, so the full tier fits a downstream model 40,800 times (50,400 on humanness) and the fast tier 4,080 times. Where a dataset leaves a downstream model out (Additional file 13), its pairs drop out: a fast search on H3K4me3 has 9 pairs, so 432 configurations.

Downstream model	Internal evaluators	Full tier	Fast tier
Random forest	3 (matched is random forest)	1,440	144
LightGBM	3 (matched is LightGBM)	1,440	144
DNN	3 (matched is DNN)	1,440	144
XGBoost	4	1,920	192
SVM	4	1,920	192
Gaussian process (humanness only)	4	1,920	not run
Total	17 pairs (21 on humanness)	8,160 (10,080 on humanness)	816

Additional file 15: Design experiments behind the search settings

One row is one setting of the search, or one rule that was tested and then fixed instead of searched. Every number comes from preliminary design runs made before the main experiments; they are not results of the thesis, and several used other budgets than the final search.

Table S.15: Design experiments behind the search settings. Preliminary design runs, not the thesis's full runs.

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Downstream model	Nine models, among them random forest, LightGBM, XGBoost, CatBoost, DNN, k-nearest neighbors and a Gaussian process, each on each of 10 base representations at full width	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, LogD, breast cancer PAM50 subtype; 5 folds	Best held-out score over the 10 representations, and seconds per fit. The DNN scored best on 4 of the 5 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio)	Full: random forest, LightGBM, XGBoost, SVM, DNN, and the Gaussian process on humanness / fast: the same five
Budget	None compared	–	–	10, 25, 50, 100, 250, 500, 1,000 and 2,000 features at both tiers
Representation count	2, 3, 5 and all 10	9 datasets, 3 tree-ensemble downstream models, budgets 500 and 1,000 features; 5 folds	The best count for each (downstream model, budget) combination	Full: 2, 3, 5, all / fast: 3, all

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Allocation method	8 candidates: the five searched and 3 further pooled candidates, each at every representation count	The same 9 datasets, models and budgets: 54 (dataset, model, budget) combinations	Each candidate was scored at its best representation count, and the number of combinations in which it was in the top 3 of the 8 was counted. Fixed shares: linear 30, squared 27, rank decay 26, softmax 25; global pool with squared weights 14	Full: the five described in Chapter 5 / fast: linear, squared, rank decay
Importance method, agreement	8 methods	LogD, hERG K_i, Buchwald–Hartwig yield, nanobody thermostability, antibody FcRn retention, breast cancer PAM50 subtype, lung adenocarcinoma vs squamous; each method fitted once on the whole dataset	Spearman correlation between two methods' scores over every feature. Kept pairs: LightGBM gain and TreeSHAP 0.20–0.64, LightGBM gain and DeepLIFT 0.21–0.67, TreeSHAP and DeepLIFT 0.56–0.99 across the 7 datasets	One method kept from each group whose rankings agree above 0.95
Importance method, score	28 methods, among them three controls: feature variance, Laplacian score and random ranking	Caco-2 efflux; 3 folds; budget 100 features; random forest and LightGBM downstream	RMSE of the best single representation minus RMSE of the hybrid ($\log_{10}$ efflux ratio; positive means the hybrid has the lower error): DeepLIFT +0.0144, LightGBM gain +0.0100, TreeSHAP +0.0067, random ranking −0.0118	Full: LightGBM gain, TreeSHAP, DeepLIFT / fast: LightGBM gain

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Internal evaluator settings	20 hyperparameter settings from 5 model families	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression; 3 folds; budget 100 features	Spread of mean scores across all 20 settings: 0.0253 R^2, 0.0179 R^2 and 0.0085 Spearman. Median standard deviation across the 3 folds for one setting, allocation method and downstream model: 0.0462 R^2, 0.0693 R^2 and 0.0186 Spearman	Matched, random forest, LightGBM and DNN, at fixed settings, both tiers
Rescaling of each representation's score (a rule)	5 rescalings	9 datasets; 5 folds; budgets 50 and 300 features	Whether any rescaling beat the one in use by more than that rule's own spread across folds: in 0 of 90 (dataset, allocation method, budget) combinations	One fixed rule per metric (Chapter 5)

The representation-count and allocation rows come from one design run on 10 datasets, 3 downstream models and 2 budgets (60 combinations); QM9 is left out, which gives the 54.

Additional file 16: Fields a blueprint saves

One row is one field. The first block is the recipe that rebuilds a hybrid; the second is kept for inspection and is not read by a rebuild.

Table S.16: Fields a blueprint saves. The stored score of each representation is its best score on its own across the downstream models, over every molecule the search ran on. It is not the internal-evaluator score that splits the budget, which a rebuild computes again on the molecules it is given. Three further fields are read only by allocation or augmentation routes outside the thesis's search and are fixed: a minimum improvement a whole block of extra features must show (none), a number of features taken from each source (20), and a step size (5).

Field	What it holds
The recipe	
Representations	The representations offered to the search
Representation count	How many of them the configuration used; empty when it used all
Allocation method	One of the five described in Chapter 5

Field	What it holds
Importance method	LightGBM gain, TreeSHAP or DeepLIFT
Budget	Number of features in the hybrid
Internal evaluator	Matched, random forest, LightGBM or DNN
Objective	Predictive, or predicting each molecule's activity from its five nearest neighbors
Metric	The dataset's metric, and whether lower is better
Task type	Regression or classification
Folds and seed	The number of folds, and the random seed
The recipe: augmentation	
Augmentation setting	None, uncapped or capped
Candidate ranking	How candidate extra features are ranked
Candidate list length	50 features
Cap per source	Optional; the most candidates taken from one source
Margin	How much better than adding none a block of extra features must score
Fraction of the budget	Optional; used instead of the fixed block sizes
Kept for inspection, not read by a rebuild	
Representation scores	Each representation's best score on its own across the downstream models, sign flipped for metrics where lower is better
Features per representation	The number of features each representation got in each fold
Extra features kept	The extra features augmentation kept in each fold, by source

S.4 Hybrid representations for similarity and label noise

Additional file 17: Settings of the twelve LNL methods and the plain network they share

One row is one method; the first row is the plain network every method starts from. The kind of each method follows Chapter 6.

Table S.17: Settings of the twelve LNL methods. On classification datasets the squared-error loss is replaced by cross-entropy.

Method	Kind	Settings	Labels
Plain network	–	Hidden layers of 256, 128 and 64 units, ReLU, dropout 0.2; Adam at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; gradient norm clipped at 1; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules held out by scaffold group (drawn at random after a method has removed molecules); labels z-normalized	both
Small-loss selection	noise mitigation	10 warm-up epochs on all molecules; keeps the 70% with the smallest loss and fits a new plain network on them	both
Confident learning	noise mitigation	3 plain networks; removes molecules whose error is above the 85th percentile and whose spread across the 3 networks is below the 25th percentile, then refits	both
Removal by distance to neighbors	noise mitigation	10 nearest neighbors in the standardized features; score is the distance of a label from its neighbors' mean label, divided by their standard deviation (on classification, the share of neighbors with another label); removes molecules above the 85th percentile, then refits	both
Staged removal	noise mitigation	3 stages; each removes the 10% of the remaining molecules with the largest training loss	both
Per-epoch down-weighting	noise-robust training	After each epoch, the weight of every molecule whose loss is above the median is multiplied by 0.95	both
Curriculum by confidence	noise-robust training	20 warm-up epochs, then trains on the 25%, 50%, 75% and 100% of molecules with the smallest loss, in stages	both

Method	Kind	Settings	Labels
DivideMix	noise-robust training	Two networks, 10 warm-up epochs; a two-component Gaussian mixture on each network's squared errors splits the molecules	both
Meta-Weight-Net	noise-robust training	The weighting network is updated every 5 epochs at learning rate 0.001, on 20% of the training molecules held out for it	both
MentorNet	noise-robust training	The mentor network reads each molecule's current loss, its last 5 losses and its mean distance to its 10 nearest neighbors	both
Contrast to Divide	noise-robust training	5 perturbed copies of each molecule, Gaussian noise of standard deviation 0.01 on the features	both
Heteroscedastic GP	noise-robust training	RBF kernel, 2 optimizer restarts, fitted on at most 1,000 training molecules; a small network predicts each molecule's noise variance	regression
Deep evidential regression	noise-robust training	Weight of the evidence penalty 0.01	regression

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